

657. Robot Return to Origin

Solved 

Easy

 Topics

 Companies

There is a robot starting at the position $(0, 0)$, the origin, on a 2D plane. Given a sequence of its moves, judge if this robot **ends up at** $(0, 0)$ after it completes its moves.

You are given a string `moves` that represents the move sequence of the robot where `moves[i]` represents its i^{th} move. Valid moves are `'R'` (right), `'L'` (left), `'U'` (up), and `'D'` (down).

Return `true` if the robot returns to the origin after it finishes all of its moves, or `false` otherwise.

Note: The way that the robot is "facing" is irrelevant. `'R'` will always make the robot move to the right once, `'L'` will always make it move left, etc. Also, assume that the magnitude of the robot's movement is the same for each move.

Example 1:

Input: `moves = "UD"`

Output: `true`

Explanation: The robot moves up once, and then down once. All moves have the same magnitude, so it ended up at the origin where it started. Therefore, we return `true`.

Example 2:

Input: `moves = "LL"`

Output: `false`

Explanation: The robot moves left twice. It ends up two "moves" to the left of the origin. We return `false` because it is not at the origin at the end of its moves.

Constraints:

- `1 <= moves.length <= 2 * 104`
- `moves` only contains the characters `'U'`, `'D'`, `'L'` and `'R'`.

Python:

class Solution:

def judgeCircle(self, moves: str) -> bool:

if len(moves) & 1: return False

x = y = 0

```
for c in moves:
    y += (c == 'U') - (c == 'D')
    x += (c == 'R') - (c == 'L')
```

```
return not x and not y
```

JavaScript:

```
const judgeCircle = moves => {
    if (moves.length & 1) return false;
    let x = 0, y = 0;

    for (const c of moves) {
        y += (c === 'U') - (c === 'D');
        x += (c === 'R') - (c === 'L');
    }

    return !x && !y;
};
```

Java:

```
class Solution {
    public boolean judgeCircle(String moves) {
        int x = 0, y = 0;

        for (int i = 0; i < moves.length(); i++) {
            char c = moves.charAt(i);

            if (c == 'U') y++;
            if (c == 'D') y--;
            if (c == 'R') x++;
            if (c == 'L') x--;
        }

        return x == 0 && y == 0;
    }
}
```